

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
Student Name:	B.VANI	Reg. No.:	192472348
Section / Batch:	CSE-AI/2024	Date:	

Code of Conduct

I B.VANI(192472348) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B.VANI

Instructions

- Read each industry scenario carefully before answering.
- Analyze the given problem from a semantic analysis perspective.
- Provide structured and logical answers with proper justification.
- Support your analysis using examples from the given scenario wherever applicable.
- Use suitable diagrams, semantic representations, logical predicates, workflows, architectures, or tables whenever necessary.
- Clearly state assumptions if additional information is required.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic interpretation of user queries, identify ambiguity in meaning representation, evaluate semantic processing limitations, and improve understanding in banking chatbot applications.	Q1
Syntax-Driven Semantic Analysis	Analyze multiple interpretations of spoken commands, evaluate parsing-related ambiguity, and assess semantic extraction in real-time voice assistant systems.	Q2
First-Order Predicate Calculus, Representing Linguistically Relevant Concepts, Syntax-Driven Semantic Analysis	Design a semantic processing architecture for healthcare reports, model relationships among entities and actions, represent linguistic concepts, and improve semantic extraction accuracy.	Q3

Annexure A – Industry Problem Solving Task

1. AI-Powered Insurance Claim Processing System

An insurance company is developing an NLP-based system to automatically process customer insurance claims. The system uses **Context-Free Grammar (CFG)** to parse customer statements and extract information such as the claimant, damaged object, cause of damage, and claim amount.

However, the system faces the following problems:

- Ambiguous sentence structures
- Difficulty distinguishing noun and prepositional phrase attachment
- Subject–verb agreement errors
- Inefficient parsing of lengthy customer descriptions
- Difficulty handling conversational and incomplete input

Example customer statement:

"I reported the damage to the car after the accident."

The sentence can produce different interpretations depending on whether "**to the car**" is associated with *reported* or *damage*.

Tasks:

- a) Analyze the structural ambiguity in the given sentence using CFG and illustrate the possible interpretations.
- b) Evaluate the limitations of a basic CFG in handling ambiguity, agreement, and long customer-generated insurance statements.
- c) Design an improved parsing approach using **PCFG, Feature Structures, and Earley Parsing** to resolve ambiguity and efficiently process the claim.
- d) Justify how the proposed architecture can improve the accuracy, scalability, and reliability of automated insurance claim processing.

2. Intelligent Railway Ticket Booking Assistant

A railway company is developing a conversational AI assistant for ticket booking. The system currently uses **top-down parsing** to interpret passenger requests.

The system encounters:

- Excessive backtracking
- Ambiguous attachment of phrases
- Incomplete passenger requests
- Long-distance dependencies
- Slow response for complex queries

Example passenger request:

"Book two tickets from Chennai to Delhi for my parents with AC sleeper."

The system sometimes incorrectly associates "**with AC sleeper**" with the passengers instead of the ticket/class information.

Tasks:

- a) Analyze the possible syntactic interpretations of the given passenger request using CFG.
- b) Evaluate why top-down parsing may result in excessive backtracking and inefficient processing for conversational ticket-booking queries.
- c) Analyze how **Earley Parsing** can handle ambiguity, incomplete input, and different possible parse structures more effectively.
- d) Compare **Top-Down Parsing, CFG-based parsing, and Earley Parsing** in terms of ambiguity handling, computational efficiency, partial-input processing, and suitability for real-time railway booking systems.

3. AI-Based Legal Contract Analysis System

A legal technology company is developing an NLP system to analyze contracts and automatically identify:

- Parties involved
- Obligations

- Conditions
- Deadlines
- Actions
- Exceptions

The existing system uses a basic CFG parser. However, it produces incorrect interpretations when processing long legal sentences containing relative clauses, passive constructions, and multiple prepositional phrases.

Example contract sentence:

"The supplier who delivered the equipment to the company must replace the defective components within thirty days."

The system sometimes incorrectly identifies the **company** as the entity responsible for delivering the equipment.

Tasks:

- Analyze the syntactic structure of the given sentence using CFG and identify possible sources of ambiguity.
- Evaluate why basic CFG is insufficient for accurately interpreting complex legal sentences containing relative clauses and nested structures.
- Design an NLP architecture integrating **CFG, PCFG, Feature Structures, and Earley Parsing** to improve syntactic and semantic interpretation.
- Develop a step-by-step processing workflow showing how the system can extract:
 - Supplier
 - Action
 - Equipment
 - Company
 - Obligation
 - Deadline

and justify how the proposed approach improves automated contract analysis.

4. AI-Based E-Commerce Product Search System

An e-commerce company is developing a conversational product-search system. Customers enter natural-language queries rather than selecting predefined filters.

The current system uses CFG but faces difficulties with:

- Ambiguous phrase attachment
- Coordination
- Missing words
- Informal customer queries
- Long product descriptions
- Real-time search requirements

Example user query:

"Find laptops with a touchscreen for students under ₹60,000."

The system may incorrectly associate "**for students**" with the touchscreen rather than the laptop.

Tasks:

- Analyze the syntactic ambiguity in the given query and construct possible parse interpretations using CFG.
- Evaluate the limitations of CFG in handling informal, elliptical, and ambiguous e-commerce search queries.
- Propose an improved parsing architecture using **PCFG, Feature Structures, and Earley Parsing** to identify product attributes, user requirements, and price constraints.
- Justify how the proposed solution can improve search precision, response time, and customer experience in a large-scale e-commerce platform.

5. Intelligent Airline Voice Assistant

An airline is developing a real-time voice assistant that allows passengers to modify flight bookings using spoken commands.

The system currently uses a conventional top-down parser. It experiences:

- Backtracking
- Ambiguous attachment
- Speech recognition errors
- Incomplete commands
- Delayed responses
- Difficulty interpreting corrections made by users

Example spoken command:

"Change my flight to Mumbai tomorrow with two passengers."

Depending on the parse, "**with two passengers**" may be incorrectly interpreted as an attribute of the destination rather than the number of passengers.

The passenger may also say:

"Change my flight to Mumbai tomorrow... actually, make that Bangalore."

Tasks:

- Analyze the possible syntactic structures and ambiguities in the given voice command.
- Evaluate the limitations of top-down parsing when processing incomplete, corrected, and ambiguous spoken commands in real time.
- Design a parsing architecture using **Earley Parsing, PCFG, and Feature Structures** to support partial input, ambiguity resolution, and agreement constraints.
- Develop a processing workflow showing how speech input can be converted into a structured booking request and justify the proposed method for real-time airline applications.

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

		performance evaluation			
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Q. No. / Criteria Level	Understanding of Industry Problem	Application of Engineering Concepts	Solution Design	Use of Tools	Analysis	Professionalism	Sub. Total	Total %
1								
2								
3								
4								
5								

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

DSA0306-NATURAL LANGUAGE PROCESSING

CO4 AT 2

B.VANI(192472348)

Q1. AI-POWERED INSURANCE CLAIM PROCESSING SYSTEM

a) Analyze the structural ambiguity using CFG.

Sentence:

"I reported the damage to the car after the accident."

A basic CFG can represent the sentence as:

$S \rightarrow NP VP$

$NP \rightarrow \text{Pronoun}$

$VP \rightarrow V NP PP PP$

$NP \rightarrow \text{Det } N$

$PP \rightarrow P NP$

Possible Interpretation 1:

I reported [the damage to the car] [after the accident].

Here, "to the car" modifies "damage".

Meaning: The damage was to the car, and it was reported after the accident.

Semantic representation:

Report(Agent=I, Object=Damage, Location=Car, Time=AfterAccident)

Possible Interpretation 2:

I reported [the damage] [to the car] [after the accident].

Here, "to the car" is interpreted as being associated with the reporting action.

Meaning: I reported the damage to the car, and the reporting happened after the accident.

The ambiguity occurs because the prepositional phrase "to the car" can attach to the noun phrase "the damage" or to the verb phrase "reported".

b) Limitations of a basic CFG.

1. CFG can generate multiple valid parse trees for an ambiguous sentence.
 2. It does not assign probabilities to competing interpretations.
 3. Basic CFG cannot naturally enforce subject-verb agreement such as "The customer reports" versus "The customers report".
 4. Large and lengthy customer descriptions can produce many possible parse trees, increasing computational cost.
 5. Conversational and incomplete inputs such as "car damaged... need claim" may not fit the predefined grammar.
 6. Domain-specific insurance terminology may not be adequately represented.
- Therefore, a basic CFG is insufficient for reliable real-world insurance claim processing.

c) Improved parsing approach using PCFG, Feature Structures and Earley Parsing.

Proposed architecture:

Customer Input



Preprocessing



Feature Structure Analysis



Earley Parser



PCFG Probability Ranking



Best Parse Selection



Semantic Role Extraction



Claim Information

PCFG:

A Probabilistic Context-Free Grammar assigns probabilities to alternative parse trees. If two structures are possible, the structure with the higher probability can be selected.

Feature Structures:

Feature structures represent grammatical properties such as:

Number = Singular/Plural

Tense = Past/Present

Person = First/Third

They help enforce grammatical agreement.

Earley Parsing:

Earley parsing stores partial states and avoids repeated parsing work. It is suitable for ambiguous and incomplete input and can efficiently handle different possible parse structures.

d) How the architecture improves the system.

Accuracy:

PCFG ranks competing interpretations, while feature structures enforce grammatical constraints.

Scalability:

Earley parsing avoids unnecessary repeated work and can process complex sentences more efficiently.

Reliability:

Semantic-role extraction maps:

Claimant → Customer

Damaged Object → Car

Cause → Accident

Claim Amount → Extracted Amount

The combination reduces ambiguity and improves the reliability of automated insurance claim processing.

Result

The proposed PCFG + Feature Structure + Earley Parsing architecture provides better ambiguity resolution, grammatical validation, and efficient parsing for customer-generated insurance statements.

Q2. INTELLIGENT RAILWAY TICKET BOOKING ASSISTANT

a) Analyze the possible syntactic interpretations using CFG.

Sentence:

"Book two tickets from Chennai to Delhi for my parents with AC sleeper."

Possible Interpretation 1:

[Book [two tickets] [from Chennai to Delhi] [for my parents] [with AC sleeper]]

Meaning:

Two AC sleeper tickets are to be booked from Chennai to Delhi for the parents.

Possible Interpretation 2:

[Book [two tickets from Chennai to Delhi] [for my parents with AC sleeper]]

Here, "with AC sleeper" can incorrectly attach to "my parents".

The intended interpretation is Interpretation 1 because "AC sleeper" describes the ticket/class rather than the passengers.

Semantic representation:

Book(

Tickets=2,

Source=Chennai,

Destination=Delhi,

Passengers=Parents,

Class=AC Sleeper

)

b) Why can top-down parsing cause excessive backtracking?

Top-down parsing starts from the start symbol and predicts grammar structures before seeing all the input.

Problems:

1. Several grammar rules may match the same input.
2. Ambiguous phrase attachment causes multiple alternatives.
3. Wrong choices must be abandoned through backtracking.
4. Long queries create more possible parse paths.
5. Incomplete conversational input can cause repeated failed predictions.
6. Repeated backtracking increases response time.

Therefore, top-down parsing may be inefficient for real-time ticket booking.

c) How Earley Parsing handles the problem.

Earley Parsing uses three operations:

Prediction

Scanning

Completion

It stores partial parse states and avoids repeatedly rebuilding the same structures.
For the query:

Book two tickets from Chennai to Delhi for my parents with AC sleeper.

Earley parsing can maintain multiple interpretations simultaneously and delay the final decision until enough context is available. It can also process partial input such as:

Book two tickets from Chennai...

and continue as more words arrive.

Thus, Earley parsing is more suitable for ambiguous and incomplete conversational input.

d) Comparison.

Feature	Top-Down Parsing	CFG-Based Parsing	Earley Parsing
Ambiguity	Poor to Moderate	Can represent ambiguity	Handles ambiguity well
Backtracking	High	Depends on parser	Reduced through chart parsing
Efficiency	Lower for complex input	Moderate	Good for general CFGs
Partial Input	Limited	Limited	Good
Long Queries	Less suitable	Moderate	Suitable

Real-Time Use	Less suitable	Moderate	More suitable
------------------	---------------	----------	---------------

Result

Earley parsing is more appropriate for railway booking because it can handle ambiguity, partial input, and complex query structures while reducing unnecessary repeated parsing.

Q3. AI-BASED LEGAL CONTRACT ANALYSIS SYSTEM

a) Analyze the syntactic structure using CFG.

Sentence:

"The supplier who delivered the equipment to the company must replace the defective components within thirty days."

Basic structure:

$S \rightarrow NP VP$

$NP \rightarrow Det N RelClause$

$RelClause \rightarrow who VP$

$VP \rightarrow V NP PP$

$VP \rightarrow V NP PP$

Simplified structure:

S

|— NP

| |— Det → The

| |— N → supplier

| |— Relative Clause

| |— who

| |— delivered the equipment to the company

|— VP

- |— must replace
- |— the defective components
- └— within thirty days

Sources of ambiguity:

1. The relative clause "who delivered the equipment to the company" is nested inside the subject NP.
 2. The PP "to the company" can create attachment ambiguity.
 3. The sentence contains multiple noun phrases and prepositional phrases.
 4. The relative pronoun "who" refers to "the supplier", not "the company".
- The correct interpretation is:

Supplier → delivered equipment

Company → Recipient

Supplier → replace defective components

Deadline → within thirty days

b) Why is basic CFG insufficient?

1. Basic CFG can produce multiple valid parse structures.
 2. It does not inherently identify the correct semantic relation.
 3. Relative clauses require correct antecedent resolution.
 4. Nested structures increase parsing complexity.
 5. Passive constructions can create subject-agent ambiguity.
 6. Multiple prepositional phrases can attach to different constituents.
 7. CFG does not assign probabilities to competing interpretations.
- Therefore, a basic CFG alone cannot reliably perform complex legal interpretation.

c) Improved NLP architecture.

Legal Document

↓

Sentence Segmentation

↓

Tokenization

↓

CFG Parsing



Feature Structure Analysis



PCFG Probability Ranking



Earley Parsing



Dependency/Semantic Analysis



Entity and Obligation Extraction



Structured Contract Information

CFG provides grammatical structure.

PCFG ranks alternative parses.

Feature Structures enforce agreement and grammatical constraints.

Earley Parsing handles ambiguity and complex structures efficiently.

Semantic analysis extracts parties, actions, conditions, deadlines, and exceptions.

d) Step-by-step extraction workflow.

Input:

"The supplier who delivered the equipment to the company must replace the defective components within thirty days."

Step 1: Tokenization

The | supplier | who | delivered | the | equipment | to | the | company | must | replace |
the | defective | components | within | thirty | days

Step 2: Identify entities:

Supplier

Equipment

Company

Defective Components

Step 3: Analyze relative clause:

who delivered the equipment to the company

"who" refers to "supplier".

Therefore:

Supplier = Deliverer

Equipment = Object

Company = Recipient

Step 4: Identify obligation:

must replace the defective components

Therefore:

Obligated Party = Supplier

Action = Replace

Object = Defective Components

Step 5: Identify deadline:

within thirty days

Therefore:

Deadline = 30 days

Final semantic representation:

Supplier → delivered → Equipment

Supplier → delivered Equipment → Company

Supplier → must replace → Defective Components

Deadline → 30 days

Final Extracted Information

Information	Extracted Value
-------------	-----------------

Supplier	The supplier	
Action	Replace	
Equipment	The equipment	
Company	The company	
Obligation	Replace components	defective
Deadline	Within thirty days	

Result

The integrated CFG, PCFG, Feature Structure, and Earley Parsing approach correctly handles nested relative clauses, PP attachment, semantic roles, obligations, and deadlines, making it suitable for automated contract analysis.

Q4. AI-BASED E-COMMERCE PRODUCT SEARCH SYSTEM

a) Analyze the syntactic ambiguity using CFG.

Query:

"Find laptops with a touchscreen for students under ₹60,000."

Possible Interpretation 1:

Find [laptops [with a touchscreen] [for students] [under ₹60,000]]

Meaning:

Find laptops that have touchscreen functionality, are suitable for students, and cost less than ₹60,000.

Possible Interpretation 2:

Find [laptops [with [a touchscreen for students]] [under ₹60,000]]

Here, "for students" is incorrectly attached to "touchscreen".

The intended interpretation is Interpretation 1.

Semantic representation:

Search(

Product=Laptops,

Attribute=Touchscreen,

UserRequirement=Students,

Price<₹60,000

)

b) Limitations of CFG for e-commerce queries.

1. CFG may generate multiple interpretations for ambiguous phrase attachment.
2. Informal queries may contain incomplete grammar.
3. Elliptical queries such as "laptops under 60k touchscreen" may not follow complete CFG rules.
4. Missing words can make syntactic parsing difficult.
5. Long product descriptions can generate many possible structures.
6. CFG does not use probabilities to determine the most likely interpretation.
7. Real-time systems require fast parsing, which can be difficult with complex grammar rules.

c) Improved parsing architecture.

User Query



Text Preprocessing



Earley Parsing



PCFG Ranking



Feature Structure Matching



Semantic Role/Attribute Extraction

↓

Search Constraints

↓

Product Retrieval

Feature structure:

Product = Laptop

Touchscreen = Yes

TargetUser = Student

MaximumPrice = ₹60,000

PCFG selects the most probable syntactic interpretation.

Earley Parsing handles incomplete and ambiguous queries.

Feature Structures ensure that extracted attributes are assigned to the correct entities.

d) Benefits.

Search Precision:

Correctly associates "for students" with "laptops" rather than "touchscreen".

Response Time:

Earley parsing manages partial and complex queries efficiently.

Customer Experience:

Users can enter natural-language requests without selecting predefined filters.

Scalability:

The architecture can handle different products, attributes, prices, and user requirements.

Final structured request:

Product = Laptops

Touchscreen = Yes

Target = Students

Price $\leq$ ₹60,000

Result

The proposed architecture improves search precision by correctly identifying relationships between products and attributes. PCFG resolves ambiguity, Feature

Structures represent constraints, and Earley Parsing supports efficient processing of natural-language queries.

Q5. INTELLIGENT AIRLINE VOICE ASSISTANT

a) Analyze possible syntactic structures and ambiguities.

Command:

"Change my flight to Mumbai tomorrow with two passengers."

Possible Interpretation 1:

Change [my flight] [to Mumbai] [tomorrow] [with two passengers].

Meaning:

Change the flight destination to Mumbai, for tomorrow, for two passengers.

Possible Interpretation 2:

Change [my flight to Mumbai] [tomorrow with two passengers].

Here, "with two passengers" may be incorrectly attached to the destination/flight phrase.

The intended semantic representation is:

ChangeFlight(

Flight=MyFlight,

Destination=Mumbai,

Date=Tomorrow,

Passengers=2

)

Second command:

"Change my flight to Mumbai tomorrow... actually, make that Bangalore."

The user corrects the destination from Mumbai to Bangalore.

The system must update:

Destination: Mumbai → Bangalore

while preserving:

Date = Tomorrow

Passengers = 2

b) Limitations of top-down parsing.

1. It can require extensive backtracking when several parses are possible.
 2. Spoken input may be incomplete.
 3. Speech recognition errors can create unexpected tokens.
 4. Corrections can invalidate an earlier interpretation.
 5. Ambiguous phrase attachment can produce incorrect structures.
 6. Repeated parsing can increase response time.
 7. Conventional top-down parsers are not ideal for continuously arriving speech.
- Therefore, top-down parsing may not satisfy strict real-time airline requirements.

c) Improved parsing architecture.

Speech Input



Speech Recognition



Tokenization



Earley Parsing



PCFG Ranking



Feature Structure Validation



Context and Correction Handling



Semantic Extraction



Booking Request

Earley Parsing:

Handles partial input and maintains possible parse states.

PCFG:

Ranks ambiguous interpretations using probabilities.

Feature Structures:

Represent constraints such as:

Passengers = 2

Date = Tomorrow

Destination = Mumbai

Correction handling:

When the user says:

Actually, make that Bangalore.

the system updates the destination feature:

Destination: Mumbai → Bangalore

d) Processing workflow.

Step 1: Speech recognition converts spoken audio into text.

"Change my flight to Mumbai tomorrow with two passengers."

Step 2: Tokenization:

Change | my | flight | to | Mumbai | tomorrow | with | two | passengers

Step 3: Earley Parsing creates possible parse structures.

Step 4: PCFG ranks the alternatives.

Step 5: Feature Structures extract:

Action = Change

Flight = My Flight

Destination = Mumbai

Date = Tomorrow

Passengers = 2

Step 6: Semantic validation checks whether each value has the correct role.

Step 7: User correction:

"Actually, make that Bangalore."

Step 8: Context manager updates:

Destination = Bangalore

Step 9: Final structured booking request:

```
{  
  Action: ChangeFlight,  
  Destination: Bangalore,  
  Date: Tomorrow,  
  Passengers: 2  
}
```

Architecture

Speech



ASR



Text Preprocessing



Earley Parser



PCFG Ambiguity Resolution



Feature Structure Validation



Context/Correction Resolution



Semantic Representation



Booking System

Justification

The proposed method is suitable for real-time airline applications because Earley Parsing supports partial input and ambiguity, PCFG selects the most probable interpretation, and Feature Structures enforce constraints and preserve consistent booking information. Correction handling allows the system to update only the changed value without reinterpreting the entire command.

Result

The proposed Earley + PCFG + Feature Structure architecture provides efficient parsing, ambiguity resolution, partial-input handling, and correction management for real-time airline voice assistants.